

FULL NAME: _____

BLOCK: _____

Multiplicity and Factored-Form Graphing

Lesson 3-5 · Period 2 · Student Packet

OBJECTIVES

Math Objective	Use the multiplicity of each zero to determine whether the graph of a polynomial crosses the x-axis or is tangent to (turns back from) it at each zero.
Language Objective	Students will use the frame “The zero $x = \underline{\quad}$ has multiplicity $\underline{\quad}$, so the graph $\underline{\quad}$ the x-axis.” to describe each zero’s graph behavior.
Essential Understanding	A factored-form polynomial reveals its zeros directly. The multiplicity of each zero — odd or even — controls whether the graph crosses or is tangent to the x-axis at that zero.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students apply the multiplicity rule to four factored-form polynomials, one per practice item. Every item is drawn from Savvas Lesson 3-5.
Essential Question	How does the multiplicity of each zero determine the graph’s behavior at that x-intercept?
Materials	Pencils; student packet; calculator (optional). No Chromebooks required for Period 2.

LAUNCH — Multiplicity Rule (teacher models on screen)

[BOOK] MULTIPLICITY RULES (keep printed on your page)

- If a factor appears ONCE or an ODD number of times, the zero has ODD multiplicity → the graph **CROSSES** the x-axis at that zero.
- If a factor appears an EVEN number of times, the zero has EVEN multiplicity → the graph is **TANGENT** to the x-axis (touches and turns back) at that zero.
- Example: $f(x) = (x - 2)^3(x + 1)^2 \rightarrow x = 2$ has multiplicity 3 (crosses); $x = -1$ has multiplicity 2 (tangent).

EXPLORE — Think-Pair-Share (35+ min)

Work with a partner. Use the multiplicity rules above for every problem.

Bank-item labels (in order):

1. Try It 2a
2. Try It 2b
3. Practice #14
4. Practice #15
5. Practice #16

Try It 2a

Describe the behavior of the graph of the function at each of its zeros: $f(x) = x(x + 4)(x - 1)^4$.

Try It 2b

Describe the behavior of the graph of the function at each of its zeros: $f(x) = (x^2 + 9)(x - 1)^5(x + 2)^2$.

Practice #14

Find the zeros of the function, and describe the behavior of the graph at each zero: $f(x) = x^3 - 8x^2 + 16x$.

Practice #15

Find the zeros of the function, and describe the behavior of the graph at each zero: $g(x) = x^3 - x^2 - 25x + 25$.

Practice #16

Find the zeros of the function, and describe the behavior of the graph at each zero: $f(x) = 9x^4 - 40x^2 + 16$.

SHARE / SUMMARY (5 min)**Sentence frames**

Pair share: “For $f(x) = \underline{\hspace{1cm}}$, the zero $x = \underline{\hspace{1cm}}$ has multiplicity $\underline{\hspace{1cm}}$, so the graph $\underline{\hspace{1cm}}$ the x-axis at that zero.”

Whole class: one pair at random reads their sentence. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO TOMORROW

If a cubic has zeros 2 , $1 + \sqrt{3}$, and $1 - \sqrt{3}$, what is the nature of its coefficients — rational, irrational, or complex? (Tuesday-to-Wednesday bridge; revisit at start of Period 3.)

EXIT

Today I learned that
 The rule I want to remember is
 One thing I’m still unsure about is

OPTIONAL REINFORCEMENT (not collected)

If you finish early or want extra practice before Wednesday. Answers on Desmos or ask for a check.

Optional #1 (Savvas Practice)

At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?

Optional #2 (Savvas Practice)

Sketch the graph of the function by finding the zeros: $f(x) = 3x^3 - 9x^2 - 12x$.